Calculus II

Solution to Assignment 4

Part I. Regular Problems

1. (a)

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	xe^x	0
1	$(x+1)e^x$	1
2	$(x+2)e^x$	2
3	$(x+3)e^x$	3
4	$(x+4)e^x$	4

With $n = 0$ to 4 and $a = 0$, we get

$$\begin{aligned}\sum_{n=0}^{4} \frac{f^{(n)}(0)}{n!}(x-0)^n &= \frac{0}{0!}x^0 + \frac{1}{1!}x^1 + \frac{2}{2!}x^2 + \frac{3}{3!}x^3 + \frac{4}{4!}x^4 \\ &= x + x^2 + \frac{1}{2}x^3 + \frac{1}{6}x^4\end{aligned}$$

(b)

n	$f^{(n)}(x)$	$f^{(n)}(8)$
0	$\sqrt[3]{x}$	2
1	$\frac{1}{3x^{2/3}}$	$\frac{1}{12}$
2	$-\frac{2}{9x^{5/3}}$	$-\frac{2}{288}$
3	$\frac{10}{27x^{8/3}}$	$\frac{10}{6912}$

$$\begin{aligned}\sum_{n=0}^{3} \frac{f^{(n)}(8)}{n!}(x-8)^n =& \frac{2}{0!}(x-8)^0 + \frac{\frac{1}{12}}{1!}(x-8)^1 \\ & - \frac{\frac{2}{288}}{2!}(x-8)^2 + \frac{\frac{10}{6912}}{3!}(x-8)^3 \\ =& 2 + \frac{1}{12}(x-8) - \frac{1}{288}(x-8)^2 + \frac{5}{20,736}(x-8)^3\end{aligned}$$

2.

$$\frac{x-\sin x}{x^3} = \frac{1}{x^3}\left[x - \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)!}\right] = \frac{1}{x^3}\left[x - x - \sum_{n=1}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)!}\right] = \frac{1}{x^3}\left[-\sum_{n=0}^{\infty}\frac{(-1)^{n+1}x^{2n+3}}{(2n+3)!}\right]$$
$$= \frac{1}{x^3}\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+3}}{(2n+3)!} = \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{(2n+3)!}$$

and this series also gives the required value at $x = 0$ (namely $1/6$); $R = \infty$.

3. (a)

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	$\ln(1+x)$	0
1	$(1+x)^{-1}$	1
2	$-(1+x)^{-2}$	-1
3	$2(1+x)^{-3}$	2
4	$-6(1+x)^{-4}$	-6
5	$24(1+x)^{-5}$	24
$\vdots$	$\vdots$	$\vdots$

$$\ln(1+x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots$$
$$= 0 + x - \frac{1}{2}x^2 + \frac{2}{6}x^3 - \frac{6}{24}x^4 + \frac{24}{120}x^5 - \cdots$$
$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots = \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}x^n.$$

(b) Since

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{1}{n+1}\cdot\frac{n}{1}\right| = \lim_{n\to\infty}\frac{1}{1+1/n} = 1,$$

so $R = 1$.

(c) We know that

$$\frac{1}{1+t} = \frac{1}{1-(-t)} = 1 - t + t^2 - t^3 + \cdots, \quad t \in (-1,1).$$

According to Theorem 2.1.20, we may integrate from 0 to any number x in $(-1,1)$ to obtain the power series for $\log(1+x)$:

$$\log(1+x) = \int_0^x \frac{1}{1+t}dt = \sum_{n=0}^{\infty}\int_0^x (-1)^n t^n dt = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

We thus showed the convergence of the series to $\log(1+x)$ for any $x \in (-1,1)$. Note that $\ln(1+x)$ is a continuous function on (-1,1]. By Abel's Theorem, the equation

$$\ln(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1}$$

is actually valid on (-1,1] and

$$\ln(2) = \ln(1+1) = \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}.$$

4.

$$\lim_{x\to 0} \frac{x - \ln(1+x)}{x^2} = \lim_{x\to 0} \frac{x - \left(x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5 - \cdots\right)}{x^2} = \lim_{x\to 0} \frac{\frac{1}{2}x^2 - \frac{1}{3}x^3 + \frac{1}{4}x^4 - \frac{1}{5}x^5 + \cdots}{x^2}$$
$$= \lim_{x\to 0} \left(\frac{1}{2} - \frac{1}{3}x + \frac{1}{4}x^2 - \frac{1}{5}x^3 + \cdots\right) = \frac{1}{2}$$

since power series are continuous functions.

Part II. Group Problems

1. To find the Taylor polynomial of degree 3 with center 0 (Maclaurin polynomial) for each function and estimate the remainder, we follow these steps:

   (a) $f(x) = 1 + x + x^2 + x^3 + x^4$

   Taylor Polynomial:

   Since
   $$f(0) = 1,\ f'(0) = 1,\ f''(0) = 2,\ f'''(0) = 6,$$
   the Taylor polynomial of degree 3 for $f(x)$ is
   $$T_3(x) = 1 + x + x^2 + x^3.$$

   Remainder Estimate:

   Since $f(x)$ is a polynomial, the remainder $R_3(x)$ is simply the next term in the polynomial:
   $$R_3(x) = x^4.$$

For $x \in [-1/2, 1/2]$, the maximum value of $|R_3(x)|$ is:

$$|R_3(x)| \leq \left(\frac{1}{2}\right)^4 = \frac{1}{16}.$$

(b) $f(x) = \cos(3x)$

Taylor Polynomial:

Since

$$f(0) = 1,\ f'(0) = 0,\ f''(0) = -9,\ f'''(0) = 0,$$

the Taylor polynomial of degree 3 is:

$$T_3(x) = 1 - \frac{9x^2}{2}.$$

Remainder Estimate:

The remainder term for $\cos(3x)$ is:

$$R_3(x) = \frac{f^{(4)}(c)}{4!}x^4,$$

where c is between 0 and x and $f^{(4)}(c) = 81\cos(3c)$. Since $|\cos(3c)| \leq 1$, the maximum value of $|R_3(x)|$ is:

$$|R_3(x)| \leq \frac{81}{24}\left(\frac{1}{2}\right)^4 = \frac{81}{24} \cdot \frac{1}{16} = \frac{81}{384} = \frac{27}{128}.$$

(c) $f(x) = \tan^{-1}(x)$

Taylor Polynomial:

Since

$$f(0) = 0,\ f'(0) = 1,\ f''(0) = 0,\ f'''(0) = -2,$$

the Taylor polynomial of degree 3 is:

$$T_3(x) = x - \frac{x^3}{3}.$$

Remainder Estimate:

The remainder term for $\tan^{-1}(x)$ is:

$$R_3(x) = \frac{f^{(4)}(c)}{4!}x^4,$$

where c is between 0 and x and $f^{(4)}(c) = \frac{24c(1-c^2)}{(1+c^2)^4}$. For $x \in [-1/2, 1/2]$, the maximum value of $|R_3(x)|$ occurs at $c = 1/2$:

$$|f^{(4)}(c)| \leq \frac{24 \cdot (1/2) \cdot (1 - (1/2)^2)}{(1 + (1/2)^2)^4} = \frac{9}{(5/4)^4} = \frac{9 \cdot 256}{625} = \frac{2304}{625}.$$

Thus:

$$|R_3(x)| \leq \frac{2304}{625 \cdot 24}\left(\frac{1}{2}\right)^4 = \frac{2304}{15000} \cdot \frac{1}{16} = \frac{2304}{240000} = \frac{96}{10000} = \frac{6}{625}.$$

2. (a)

$$\lim_{x\to 0} \frac{\sin x^2 - x^2}{x^6}$$
$$= \lim_{x\to 0} \frac{\left(x^2 - \frac{x^6}{6} + \cdots\right) - x^2}{x^6} = \frac{-1}{6}$$

(b) $\lim_{x\to 0} \frac{e^{-2x}-1}{x}$

$$\lim_{x\to 0} \frac{e^{-2x} - 1}{x}$$
$$= \lim_{x\to 0} \frac{\left(1 - 2x + 2x^2 - \cdots\right) - 1}{x} = -2$$

(c)

$$\lim_{x\to 0} \frac{\cos x^2 - 1}{x^4}$$
$$= \lim_{x\to 0} \frac{\left(1 - \frac{x^4}{2!} + \frac{x^8}{4!} - \frac{x^{12}}{6!} \cdots\right) - 1}{x^4}$$
$$= -\frac{1}{2}$$